

Task:

Your task is to build a simple machine learning model for a **binary classification** problem. You must choose a dataset to help solve important **healthcare** problems. (datasets can be found online)

Build a **Decision Tree** and a **K-Nearest Neighbors (KNN)** classifier to predict the target variable based on your selected dataset.

Steps:

Data Preprocessing:

- Load the dataset and examine its structure. (**Hint:** use head, shape, info, describe, etc. methods)
- The dataset should contain only numeric features. Remove the categorical features (if any)
- Drop the irrelevant features from the dataset. Also, handle the missing values appropriately
- Split the dataset into training and testing sets. Test set size will be set as – **the last 3 digits of your ID % 40**

Model Training:

- Initialize a decision tree and a KNN model using **Scikit-learn**
- Train both models on the training set

Model Evaluation:

- Evaluate the models on the testing set using appropriate metrics (e.g., accuracy, precision, recall, F1-score)
- Mention which model performs better in terms of **F1-score**

Instructions:

- Include comments in your code to explain key steps
- Everyone should try to select different datasets
- Consider using simple visualization techniques to represent the data or the results (optional)
- Consider experimenting with the hyperparameters of decision tree and KNN to improve performance (optional)